EXP 13: CASE STUDY ON LOW-CODE LIMITATIONS

AIM

To carry out a case study on the limitations of low-code development by analyzing the Library Book Issue Management application built in Zoho Creator, and to suggest ways to reduce these limitations.

PROCEDURE

1. **Select the case**
   - The case is the **Library Book Issue Management** application developed on the Zoho Creator low-code platform.
   - Low-code platforms let applications be built with visual tools (drag-and-drop forms, reports and workflows) and only a small amount of code (Deluge).
2. **Study the application built**
   - **Data model and forms:** the Book Issue Request and Book_Catalog forms with many field types.
   - **Reports and dashboards:** list, spreadsheet, Kanban, calendar, summary reports, a dashboard and a command-center page.
   - **Automation:** form, scheduled and batch workflows, a custom function, a blueprint and email notifications using Deluge.
   - **Access control:** Librarian and Student roles and permissions, and deployment through environments.
3. **Identify the limitations observed**
   - Review each part of the application and list the points where the platform restricted the design, the cost, the performance or the control.
   - Record the observations in the table of limitations given below.
4. **Analyze the impact of each limitation**
   - Check how each limitation affects the library application in terms of flexibility, cost, scalability, security and maintenance.
   - Decide which limitations are serious for a small college library and which matter only for very large systems.
5. **Compare with traditional coding**
   - Compare low-code with custom coding in terms of speed, control, cost and skills, as shown in the comparison table.
6. **Suggest solutions**
   - Plan how each limitation can be reduced, for example by good design, Deluge functions, integrations, backups and regular reviews.
   - Prepare the conclusion about when low-code is suitable and when custom development is better.

Limitations Observed in the Case

Limitation	Observation in the library application	Possible solution
Vendor lock-in	The application runs only on Zoho Creator; the .ds export cannot be used on another platform	Keep exported backups and CSV data exports; document the design
Limited customization	Form layout, page design and report appearance are restricted to the platform's components and HTML snippets	Use pages, widgets and custom HTML where needed

Limitation	Observation in the library application	Possible solution
Plan and cost restrictions	The account is a trial; some features (such as Stage and Production environments) depend on the paid plan	Check plan features early and estimate the cost
Scalability and platform limits	Scripts and API calls have limits per execution, so heavy tasks need batch or scheduled workflows	Optimize Deluge, use batch workflows, archive old data
Dependence on Deluge	Complex logic such as fine calculation still needs Deluge scripting skills	Train developers and reuse custom functions
Limited debugging and testing	Debugging is done mainly with logs and info statements; there are few advanced testing tools	Test in Stage with sample data and keep clear logs
Security and control	Security depends on the vendor and the plan; the infrastructure cannot be changed	Use roles, least privilege and review the vendor's policies
Integration limits	Connecting with other systems depends on available connectors and APIs	Use REST APIs, invokeUrl and custom functions

Low-Code Compared with Traditional Coding

Point	Low-code (Zoho Creator)	Traditional coding
Development speed	Very fast; the application was built using visual tools	Slower; needs setup, coding and testing
Control and flexibility	Limited to the platform's features	Full control over design and logic
Skills needed	Basic skills and some Deluge	Strong programming skills
Cost	Low at the start; subscription cost grows with users and features	Higher development cost; lower platform dependence
Maintenance	Platform handles hosting and updates	Team must manage servers and updates

When Low-Code Is Suitable and When It Is Not

- **Suitable:** internal business tools, small and medium applications, quick prototypes and workflow automation such as the library system.
- **Suitable:** teams with limited programming skills that need to build and change applications quickly.
- **Not suitable:** applications that need a fully custom user interface, very high performance or complex algorithms.
- **Not suitable:** systems with strict rules about hosting location, or where moving to another platform later must be easy.

Recommendations

- Check the platform limits and the plan features before starting the project.
- Keep regular backups of the application (.ds file) and of the data (CSV or Excel).
- Keep Deluge scripts short, reusable and well documented.
- Use low-code for speed, and add custom code or integrations only where the platform is not enough.

RESULT

The case study on low-code limitations was completed successfully using the Library Book Issue Management application. Low-code made it possible to build the forms, reports, workflows and access control quickly, but limitations such as vendor lock-in, limited customization, plan and platform limits, dependence on scripting and limited debugging were identified. These can be reduced by careful design, backups, optimization and regular reviews. Low-code is well suited to small and medium business applications, while very complex or highly customized systems may need traditional development.